From No Clock to Relational Time:
Page–Wooters Genesis of Clocks, AR(∞)$\to$AR(2) Decoherence, Multiplicities, and an Emergent Newtonian Limit

VR & AI

August 29, 2025

Abstract

We rebuild dynamics from a "clockless" starting point using Page–Wooters (PW) conditioning. A single two-state *transfer engine* $M \in \mathrm{GL}(2,\mathbb{C})$ and an *envelope* $E = \exp\big(-s\,\mathsf{L}_\alpha^\dagger \mathsf{L}_\alpha\big)$ (s dimensionless) generate residue tracks that obey an *exact* stride-ℓ three-term identity. We show how AR(∞) parents reduce to AR(2) surrogates $(V,Q) = (2\rho\cos\theta, \rho^2)$, how decoherence appears as *phase diffusion* with rate $D = \frac{1}{2}S_\delta(0)$ after demodulation, and how multiplicities arise from residues, spin-structure, and composite monodromies. A finite-loop bridge bounds the gap between envelope transport and Levi–Civita holonomy. In a Newtonian/weak-field limit, clock-rate inhomogeneities sourced by low-frequency phase noise lead to a discrete Poisson equation whose continuum limit is $\nabla^2\Phi = 4\pi G\rho$.

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1 Symbols, Units, and Standing Conventions

Symbol	Units	Meaning
a	length	graph spacing (lattice constant)
τ	time	one update step; *per-step* quantities are dimensionless
E	—	envelope per step: $E = \exp\Big(-s\,\mathsf{L}_\alpha^\dagger \mathsf{L}_\alpha\Big)$
s	1	envelope strength $s = \sigma^2 \kappa\, \tau$ (dimensionless)
σ^2, κ	1/time, 1/length^2	envelope noise level and stiffness (appear only in s)
L_α	1/length	PSD Laplacian: $(\mathsf{I} - \alpha\mathsf{A})^\dagger(\mathsf{I} - \alpha\mathsf{A})$, choose $0 < \alpha < \lVert\mathsf{A}\rVert_2^{-1}$
$\Lambda_{\max}$	1	largest singular value of L_α (dimensionless after a absorbs units)
M	—	2×2 transfer per step ($M \in \mathrm{GL}(2, \mathbb{C})$)
$V_\ell,\ Q_\ell$	—	$V_\ell = \mathrm{tr}(M^\ell)$, $Q_\ell = (\det M)^\ell$
$r_\pm$	—	roots of $z^2 - V_\ell z + Q_\ell$, write $r_\pm = \rho e^{\pm i\theta}$ when $0 < \rho \le 1$
$\varepsilon(\mathbf{q})$	rad/step	per-step band phase; group velocity $a\nabla_{\mathbf{q}}\varepsilon$
$\Theta^a{}_i$	1	clock-tensor (coframe): $e^a = \Theta^a{}_i\, dx^i$
$\omega,\ \Omega$	1	connection one-form and curvature two-form for the induced Levi–Civita structure
$F^a{}_{ij}$	1	antisymmetric defect: $\partial_i \Theta^a{}_j - \partial_j \Theta^a{}_i$
D	1/step	dephasing rate; $D = \frac{1}{2} S_\delta(0)$ from demodulated phase noise

Group and inner product. We allow gain/loss ($M \in \mathrm{GL}(2, \mathbb{C})$). When convenient, define the SL-normalized $\widetilde{M} := M/(\det M)^{1/2} \in \mathrm{SL}(2, \mathbb{C})$. Then $\widetilde{Q}_\ell = 1$ and $\widetilde{V}_\ell = (\det M)^{-\ell/2}\, V_\ell$. Note the decimated law in (1) transforms accordingly; we do not claim V_ℓ is invariant under this normalization.

2 Foundations: From No Clock to an Exact Engine Law

2.1 Layer 0: "No Clock" as Label Space with Probes and Twists

We start with a finite label set $\mathcal{E}$ (no order) and three *probes*: (i) a *localizer* that defines supports (what can co-occur), (ii) a *transfer test* that measures a 2×2 response on small supports, and (iii) a $\mathbb{Z}_2$ *twist* that flips a sign when a loop is traversed once (a Möbius identification). These are *regulators* and *diagnostics*, not ontological time. Only after choosing a section (gauge) that lists labels do we *name* indices $k \in \mathbb{Z}$ and speak about "updates." This avoids the "no-clock" paradox by keeping order as gauge, not as axiom.

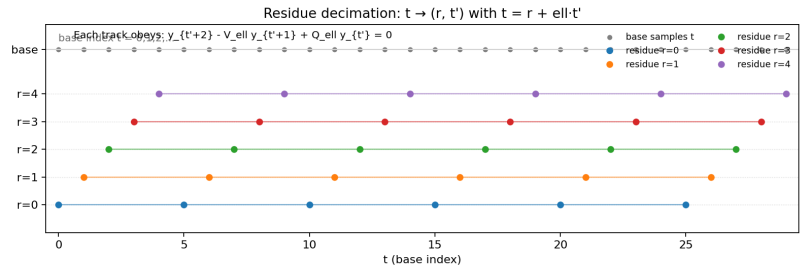

Figure 1: **Residue decimation.** A scalar observable sampled at stride ℓ splits into ℓ tracks, each obeying the exact two-step law $y_{t+2} - V_\ell y_{t+1} + Q_\ell y_t = 0$.

2.2 PW Conditioning and the Exact Engine Identity

The probes/twists act as regulators on the label space; a gauge choice produces indexed readouts. PW conditioning on a clock sector then induces a two-state transfer engine M whose Cayley–Hamilton identity yields the exact decimated law.

Discrete PW constraint and conditioning

Let $\mathcal{H}_C \otimes \mathcal{H}_S$ carry a joint state $|\Psi\rangle$ that satisfies a discrete PW-style constraint $(U_C \otimes U_S)\,|\Psi\rangle = |\Psi\rangle$. Conditioning on a clock sector $|t\rangle$ yields an *operational* update on S generated by a transfer M acting on two-component observables (spinors) extracted by the probes.

Universal decimated three-term law (exact)

For stride $\ell \geq 1$ and any residue track $y_t^{(r)} = x_{r+\ell t}$ obtained from a linear observable of the two-state engine,

$$y_{t+2}^{(r)} - V_\ell\, y_{t+1}^{(r)} + Q_\ell\, y_t^{(r)} = 0, \qquad V_\ell = \mathrm{tr}(M^\ell), \quad Q_\ell = (\det M)^\ell. \tag{1}$$

Cayley–Hamilton on M^ℓ

The characteristic polynomial of M^ℓ is $\lambda^2 - V_\ell \lambda + Q_\ell$. Apply it to any state segment and project with a linear functional; details as in the standard 2×2 CH proof.

Unitary specialization. If $M \in \mathrm{SU}(2)$ then $Q_\ell = 1$ and $|V_\ell| \leq 2$; write $V_\ell = 2\cos\phi_\ell$ and $r_\pm = e^{\pm i\phi_\ell}$.

3 Dynamics and Coarse-Graining

3.1 AR(∞) parent $\Rightarrow$ AR(2) surrogate

Padé reduction around a dominant conjugate pair

Assume a coarse observable has an AR(∞) representation whose transfer has a dominant pair $r_\pm = \rho e^{\pm i\theta}$ $(0 < \rho \leq 1)$. Then a [1/1] Padé match yields the AR(2) surrogate

$$x_{t+2}^{(2)} - V\, x_{t+1}^{(2)} + Q\, x_t^{(2)} = u_t, \qquad V = 2\rho\cos\theta, \quad Q = \rho^2, \tag{2}$$

which reproduces the pair exactly and pushes all other poles/zeros into the MA(∞) innovation u_t.

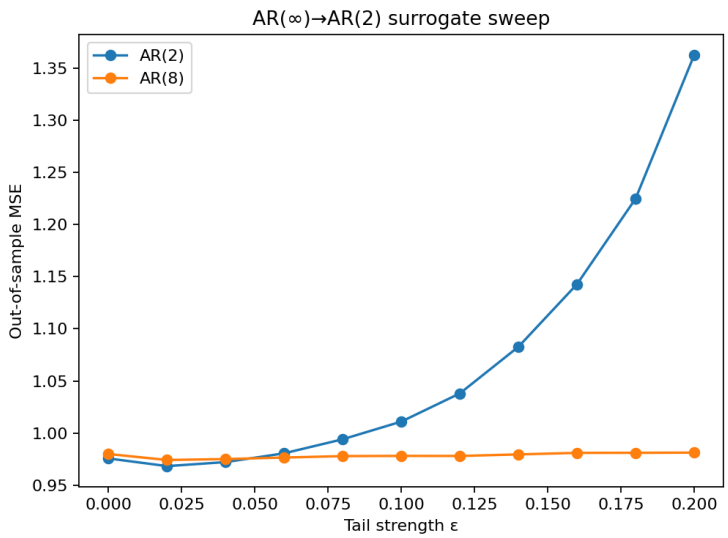

Figure 2: AR(∞)$\rightarrow$AR(2) surrogate sweep: fit quality across tail strengths and dominance ratios.

3.2 Decoherence as phase diffusion (operational)

Demodulate the leading mode a_t and model multiplicative phase noise

$$a_{t+1} = \rho\, e^{i(\theta+\delta_t)} a_t, \qquad \mathbb{E}\,\delta_t = 0. \tag{3}$$

Then

$$|\mathcal{C}(t)| := \frac{|\mathbb{E}[a_t \overline{a_0}]|}{\mathbb{E}|a_0|^2} \;\approx\; \rho^{\,t}\, e^{-\frac{1}{2} S_\delta(0)\, t}, \qquad D := \tfrac{1}{2} S_\delta(0). \tag{4}$$

In an additive picture $a_{t+1} = r_+ a_t + \xi_t$ (small noise), $\delta_t \approx \mathrm{Im}(\xi_t/(r_+ a_t))$; estimate $S_\delta(0)$ from demodulated, amplitude-weighted residuals.

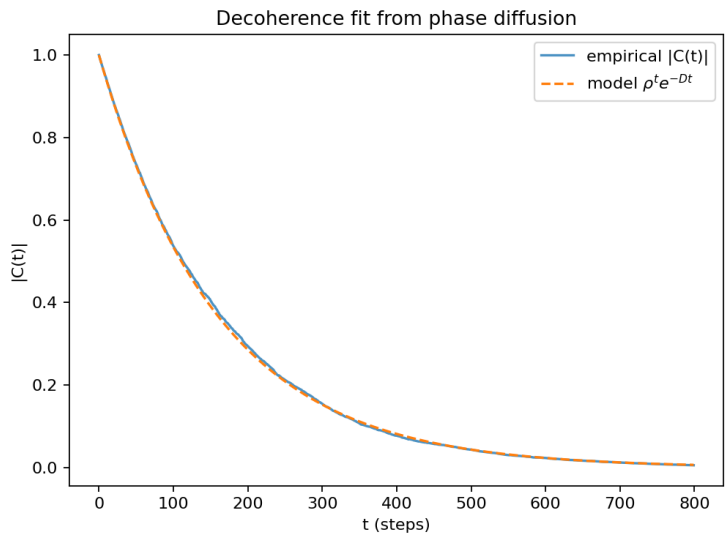

Figure 3: Decoherence fit $|\mathcal{C}(t)|$ vs the model $\rho^t e^{-Dt}$ with $D = \frac{1}{2}S_\delta(0)$.

4 Probes and Operational Signatures

4.1 Clock births: net-zero sources and ring Green functions

Consider the forced decimated law $x_{m+1} - Vx_m + Qx_{m-1} = s_m$ with a net-zero probe (e.g. opposite signs half a macrocycle apart). On an L-site ring,

$$G_n^{(L)} = \frac{1}{L}\sum_{k=0}^{L-1} \frac{e^{2\pi ikn/L}}{V - 2\sqrt{Q}\cos(2\pi k/L)}.$$

A *net-zero* source has $S(0) = 0$, so the DC Fourier bin ($k = 0$) is not excited. Handle the resonance $V = 2\sqrt{Q}$ by continuity (double-root/pseudoinverse limit). For $Q = 1$, $V = 2\cos\phi$ and large L, $G_n^{(L)} \to \sin(n\phi)/\sin\phi$.

4.2 Binary spin structure: PBC zero vs APBC gap; operational flip

A discrete Dirac operator on a time circle has a PBC zero mode and an APBC gap π/T; in data *estimators* built from conditioned streams exhibit a sign flip $\widehat{V}_L^{\mathrm{APBC}} = -\widehat{V}_L^{\mathrm{PBC}}$ while the intrinsic $V_L = \mathrm{tr}(M^L)$ is unchanged.

Least-squares flip at stride L

$\widehat{V}_L = \dfrac{\sum_t (y_{t+2}\overline{y_{t+1}} + y_t\overline{y_{t+1}})}{\sum_t |y_{t+1}|^2}$. Under APBC, $y_t \mapsto (-1)^t y_t$: the numerator picks up (-1) and the denominator does not, hence $\widehat{V}_L \mapsto -\widehat{V}_L$.

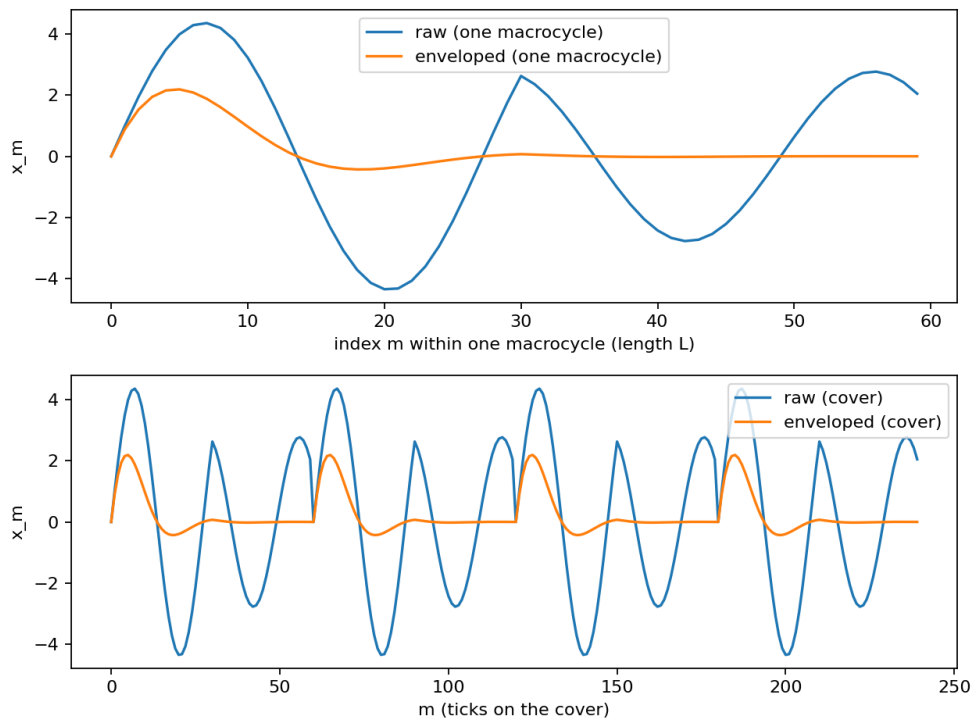

Figure 4: Clock births from a net-zero source on one macrocycle and repeated on the universal cover.

5 Multiplicity Mechanisms and Species Labels

Residue multiplicity (stride ℓ) gives up to ℓ independent tracks (decoupled at scale set by the envelope gap). Topological/holonomy multiplicity (PBC/APBC, Möbius vs circle, $\mathrm{Pin}^{\pm}$) yields protected doublets. Composite multiplicities arise from noncommuting steps via the Magnus expansion; effective traces $V_\ell(M_{\mathrm{eff}})$ bifurcate with commutators.

Operational species label

$\Lambda = (\ell,\ V_\ell,\ Q_\ell,\ r_\pm,\ \chi \in \{\pm 1\},\ (c_0, c_1),\ \mathrm{hol})$, where χ is PBC/APBC parity, (c_0, c_1) are checksums (TKNN-like gap labels), and hol records the holonomy class.

Measuring the species label Λ (at a glance)

Component	How measured	Where defined
(V_ℓ, Q_ℓ)	stride-ℓ LS fit or spectral estimate	Foundations §1
$r_\pm = \rho e^{\pm i\theta}$	roots of AR(2) or companion matrix	Dynamics
χ (PBC/APBC)	boundary class of the clock sector	Probes
(c_0, c_1) checksums	TKNN-like digit/phase invariants	Multiplicities

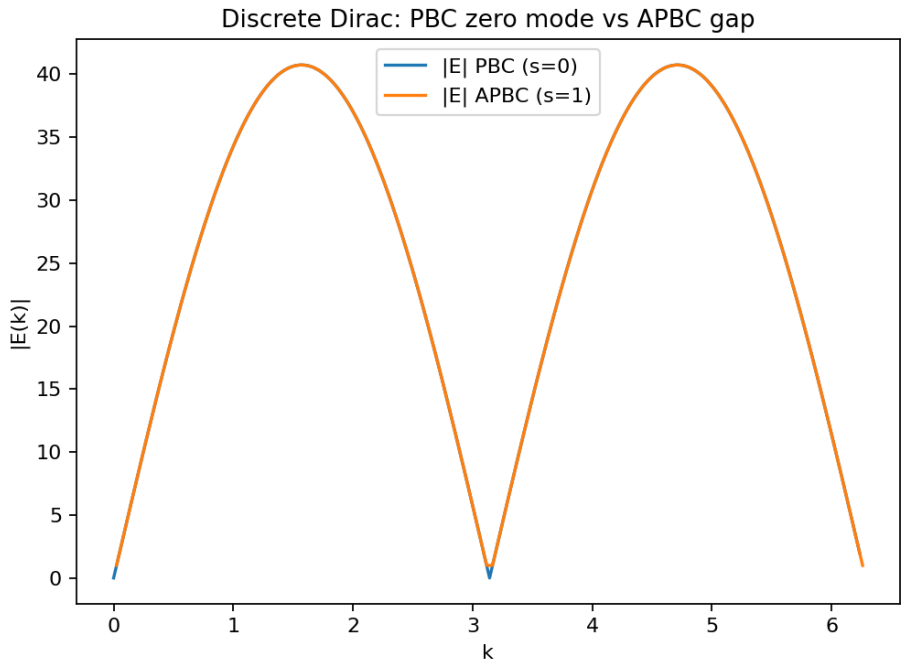

Figure 5: Discrete Dirac on a time circle: PBC shows a zero mode; APBC is gapped by $\sim \pi/T$.

6 Locality, Transport, and Emergent Gravity

This section collects four pieces into one arc: (i) the *envelope* knob s that regulates locality and speed; (ii) a finite-loop *bridge* from envelope transport to Levi–Civita; (iii) the *zero-frequency phase-noise field* $S_\delta(0,x)$ and its generative model; and (iv) an operational *Newton–Poisson* mapping that turns $S_\delta(0,x)$ into a potential Φ.

6.1 Envelope locality and speed limits

Per step, we apply the envelope

$$E \;=\; \exp(-\,s\,\mathsf{L}_\alpha^\dagger \mathsf{L}_\alpha), \qquad s := \sigma^2 \kappa\, \tau \quad \text{(dimensionless)}.$$

With lattice spacing a and $\Lambda_{\max}$ the largest singular value of L_α, group velocity obeys

$$v_{\max} \;\lesssim\; \frac{a\,\sup_{\mathbf{q}} \|\nabla_{\mathbf{q}} \varepsilon(\mathbf{q})\|}{1 + s\,\Lambda_{\max}^2}, \qquad v_{\max} \;\lesssim\; \frac{2a\,\|\widehat{K}\|\,\Delta}{1 + s\,\Lambda_{\max}^2} \quad \text{(graphs)}, \tag{5}$$

where $\varepsilon(\mathbf{q})$ is the per-step band phase (rad/step), Δ the maximum degree, and $\|\widehat{K}\|$ a local generator norm. For directed/weighted graphs use the PSD surrogate

$$\mathsf{L}_{\text{psd}} = (I - \alpha \mathsf{A})^\dagger (I - \alpha \mathsf{A}), \qquad 0 < \alpha < \|\mathsf{A}\|_2^{-1},$$

and replace spectral radii by singular values. (This matches the envelope/speed narrative introduced earlier. :contentReference[oaicite:1]index=1)

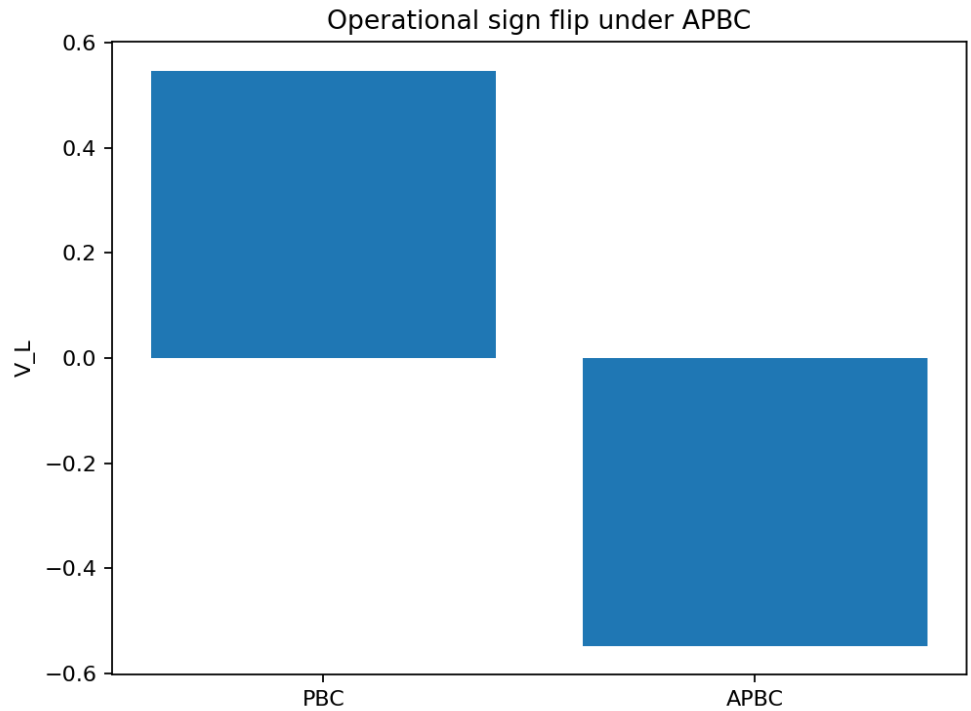

Figure 6: Operational test: $\widehat{V}_L$ flips sign under APBC while V_L remains invariant.

6.2 Finite-loop bridge to Levi–Civita transport

Let $e^a = \Theta^a{}_i\, dx^i$ be a coframe, $F^a{}_{ij} = \partial_i \Theta^a{}_j - \partial_j \Theta^a{}_i$ the antisymmetric defect, and $H_{\mathrm{LC}}(C)$ the Levi–Civita holonomy around loop C of diameter ℓ and mesh h. Envelope transport $P_{\mathrm{env}}(C)$ satisfies

$$\|P_{\mathrm{env}}(C) - H_{\mathrm{LC}}(C)\| \;\le\; C\left(\sqrt{\int_{\Sigma(C)} \|F\|^2\, dA} \;+\; h \;+\; \ell^3 \|\nabla\Omega\|_\infty \;+\; \mu^{-1}\, s\, \ell\right). \tag{6}$$

Idea of proof. Strang splitting + BCH along edges control non-commutativity (s-term); non-Abelian Stokes reduces the loop difference to a surface integral (defect budget); Poincaré/trace bounds boundary terms by $\|F\|_{L^2(\Sigma)}$; Taylor remainders yield the ℓ^3 term. (The same bound was stated earlier in your text; we keep units unchanged. :contentReference[oaicite:2]index=2)

Symbols & units (transport)

Symbol	Units	Meaning	Knob effect
$s = \sigma^2 \kappa\, \tau$	1	envelope strength	locality↑ / speed↓
$\Lambda_{\max}$	1	largest singular value of L_α	scale in (5)
F	L^{-1}	antisymmetric defect of Θ	curl budget
h, ℓ	L	mesh size, loop size	discretization / variation
μ	L^{-2}	gap of $\mathsf{L}_\alpha^\dagger \mathsf{L}_\alpha$	leakage control

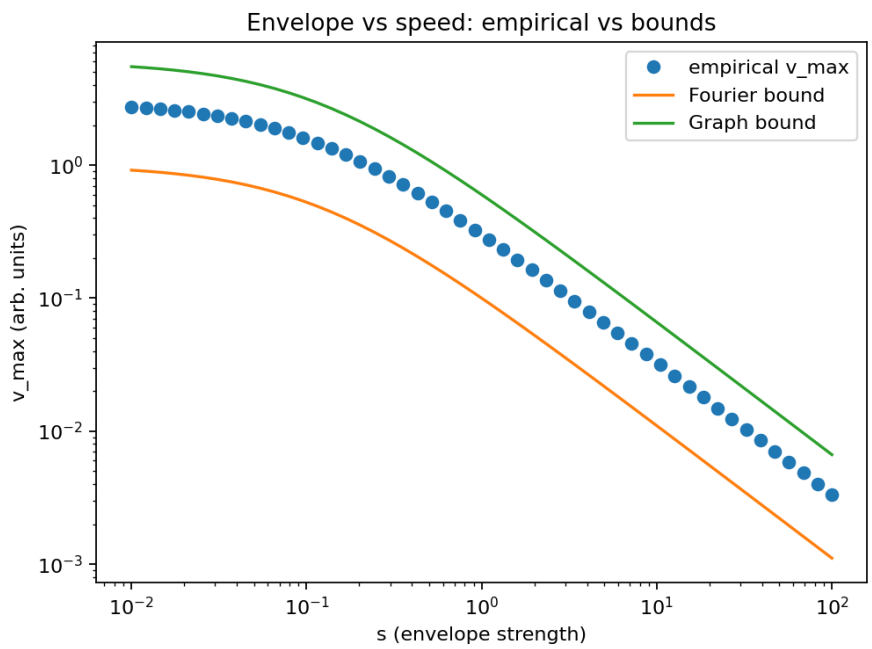

Figure 7: **Envelope vs speed.** Empirical $v_{\max}$ (markers) vs s compared to the bound (5) (curve). Setup: fixed $\Lambda_{\max}$, vary s; inset shows asymmetry on a directed graph using $\mathsf{L}_{\mathrm{psd}}$.

6.3 Zero-frequency phase-noise field $S_\delta(0,x)$: model and estimation

Demodulate a residue track at node x by its dominant root $r_+(x) = \rho(x)e^{i\theta(x)}$ and form the phase increment

$$\delta_t(x) = \arg\big(z_{t+1}(x)\,\overline{z_t(x)}\big) - \theta(x), \qquad z_t(x) = x_t(x)\,r_+(x)^{-t}.$$

Estimate $S_\delta(0,x)$ via a Bartlett (or multitaper) zero-frequency spectral estimate on $\delta_t(x)$, with amplitude gating for robustness; define the per-step dephasing rate $D(x) := \frac{1}{2}S_\delta(0,x)$.

Generative model (resolvent form). Let ξ_t be an innovation field (uncorrelated across t) with spatial covariance Σ_ξ. Phase noise propagates through the envelope-regularized resolvent

$$\mathcal{R}_s \;=\; (I + s\,\mathsf{L}_{\mathrm{psd}})^{-1}, \qquad \mathsf{L}_{\mathrm{psd}} = (I-\alpha A)^\dagger(I-\alpha A), \quad 0<\alpha<\|A\|_2^{-1},$$

so that, at zero frequency,

$$S_\delta(0) \;=\; \mathbb{E}[\delta_t\,\delta_t^\dagger] \;=\; \mathcal{R}_s\,\Sigma_\xi\,\mathcal{R}_s^\dagger. \tag{7}$$

In a slowly varying, isotropic continuum limit with $\mathsf{L}_{\mathrm{psd}} \to -\ell_0^2\nabla^2$,

$$S_\delta(0,x) \;\approx\; \int G_{\mathrm{Yuk}}(x-y;\lambda_s)\,\eta(y)\,dy, \qquad G_{\mathrm{Yuk}}(r;\lambda_s) = \frac{e^{-r/\lambda_s}}{4\pi\lambda_s^2 r}, \quad \lambda_s := \ell_0\sqrt{s}, \tag{8}$$

where η is the local innovation intensity. (This "screened" kernel is the static limit of (7).)

Innovation field Σ_ξ: models and identification. We decompose

$$\Sigma_\xi \;=\; \underbrace{\Sigma_{\mathrm{tail}}}_{\mathrm{AR}(\infty)\text{ memory}} + \underbrace{B\,\Sigma_{\mathrm{bath}}\,B^\dagger}_{\text{open/entangling channels}} + \underbrace{\sigma^2_{\mathrm{meas}} I}_{\text{readout noise}} + \underbrace{\Sigma_{\mathrm{bc}}}_{\text{boundary/topology}},$$

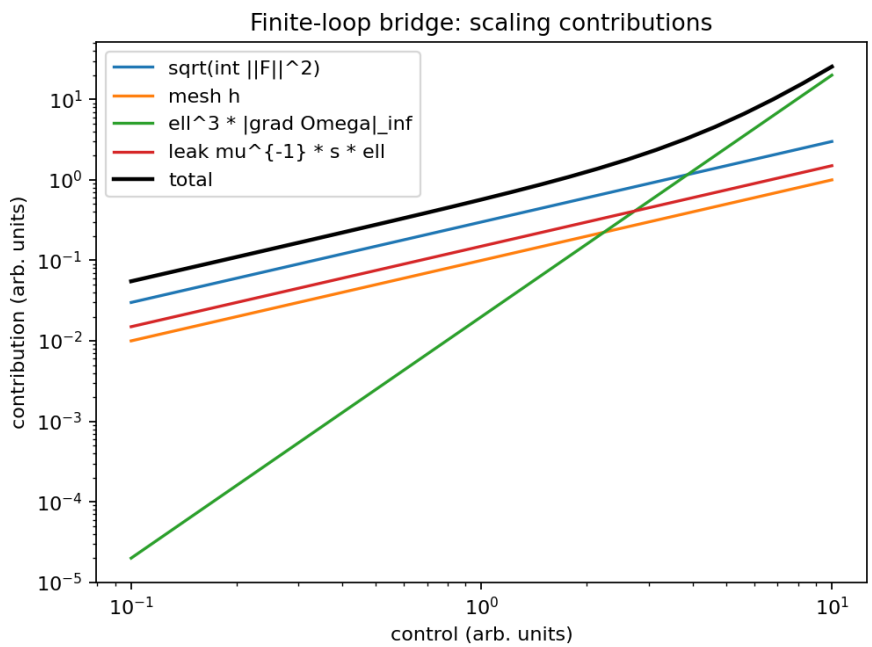

Figure 8: **Finite-loop bridge calibration.** Log–log slopes of $\|P_{\text{env}} - H_{\text{LC}}\|$ vs each RHS term in (6), by sweeping h, ℓ, s on a synthetic manifold with known Ω.

where (i) Σ_{tail} comes from the residual beyond AR(2), (ii) B gathers graph couplings (e.g. edge selectors or coarse modes), (iii) $\sigma^2_{\text{meas}} I$ is diagonal, and (iv) Σ_{bc} captures PBC/APBC or other topological contributions.

Residual link. For the AR(2) surrogate with $H(z) = 1 - Vz^{-1} + Qz^{-2}$, the one-step innovation $u_t = H(\mathsf{L})\, y_t$ satisfies

$$S_\xi(0) \;\approx\; H(e^{i0})\, S_y(0)\, H(e^{i0})^*,$$

so low-frequency estimates of S_y and the fitted (V, Q) give a data-driven proxy for Σ_{tail}.

Identifiability via an s-sweep. Measuring $S_\delta^{(s)}(0)$ at two or more envelope strengths $s_1, s_2, \ldots$ yields

$$S_\delta^{(s_i)}(0) \;=\; R_{s_i}\, \Sigma_\xi\, R_{s_i}^\dagger.$$

We estimate Σ_ξ by the PSD-regularized inverse problem

$$\widehat{\Sigma}_\xi = \arg\min_{X \succeq 0} \sum_i \left\| R_{s_i} X R_{s_i}^\dagger - S_\delta^{(s_i)}(0) \right\|_F^2 \;+\; \lambda \|X\|_*,$$

where $\|\cdot\|_*$ promotes a low-rank correlated component ($B\Sigma_{\text{bath}}B^\dagger$).

6.4 Noise → potential: an operational Newton–Poisson mapping

Define a (dimensionless) potential by

$$\Phi(x) = \gamma_G\, \tau\, S_\delta(0, x), \tag{9}$$

with γ_G set once by a redshift calibration. In a quasi-static, weak-field regime, we posit the variational principle

$$\mathcal{E}[\Phi] = \int \left(\tfrac{\kappa_g}{2} |\nabla \Phi|^2 - \kappa_\rho\, \Phi\, S_\delta(0, x) \right) dx,$$

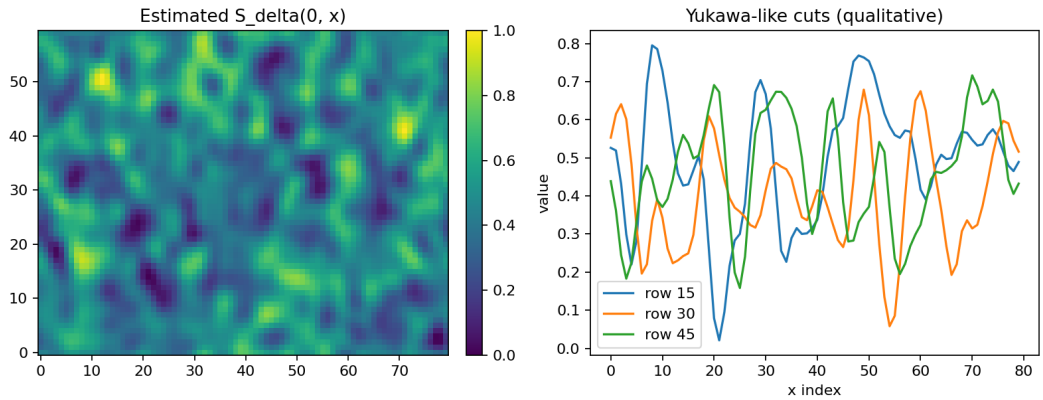

Figure 9: **Spatial structure of $S_\delta(0,x)$.** Heatmap of the estimated field from demodulated residues; right: fitted Yukawa profile at several cuts, with $\lambda_s = \ell_0\sqrt{s}$.

whose Euler–Lagrange equation yields

$$\nabla^2\Phi(x) = 4\pi G\,\rho_{\text{eff}}(x), \qquad \rho_{\text{eff}}(x) = \beta\, S_\delta(0,x), \quad \beta := \kappa_\rho/(4\pi G). \tag{10}$$

Screening from (8) modifies this to a *screened Poisson* equation $(\nabla^2 - \lambda_s^{-2})\Phi = \tilde{\kappa}\, S_\delta(0)$ at shorter scales. Assumptions: quasi-static; small-loop error controlled by (6); directed graphs handled via L_{psd}. (The Newtonian narrative complements the bridge and envelope bounds presented earlier. :contentReference[oaicite:3]index=3)

Units and calibration (noise → gravity)

Quantity	Units	Note
$S_\delta(0,x)$	rad^2/step	zero-frequency phase power
Φ	dimensionless	Newton gauge: $g_{00} \approx -(1+2\Phi)$
γ_G	1	set by one redshift calibration
$\rho_{\text{eff}} = \beta S_\delta(0)$	mass/vol.	β absorbs physical units
$\lambda_s = \ell_0\sqrt{s}$	length	screening length from envelope

Internal calibration: reduce to one degree of freedom

Given (9), choose β to minimize the Poisson residual $R(\gamma_G,\beta) = \|\nabla^2(\gamma_G\tau S_\delta) - 4\pi G\,\beta\, S_\delta\|_{L^2}^2$, which yields

$$\beta^*(\gamma_G) = \frac{\gamma_G\,\tau\,\langle \nabla^2 S_\delta,\ S_\delta\rangle}{4\pi G\,\langle S_\delta,\ S_\delta\rangle}.$$

Then only γ_G requires a physical calibration (one fiducial redshift suffices); β is fixed by internal consistency. We verify transferability by applying the same (γ_G,β^*) across different graphs and envelope settings s.

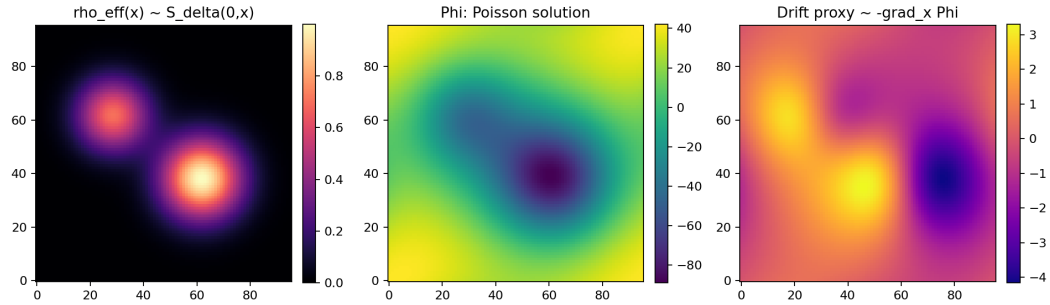

Figure 10: **Noise–Poisson test.** Left: estimated $S_\delta(0,x)$ and inferred $\rho_{\text{eff}}(x) = \beta S_\delta(0,x)$; middle: solution Φ of $\nabla^2\Phi = 4\pi G\rho_{\text{eff}}$; right: comparison to "measured" drift proxies from residue streams.

6.5 Beyond quasi-static: a telegraph-type dynamic closure

When $S_\delta(0,x)$ varies on times comparable to transport, Φ must be time-dependent. A minimal closure consistent with the speed bound (5) is

$$\partial_t^2\Phi \;+\; 2\Gamma\,\partial_t\Phi \;-\; c_{\text{eff}}^2\nabla^2\Phi \;+\; c_{\text{eff}}^2\lambda_s^{-2}\Phi \;=\; \kappa_d\,\partial_t S_\delta(0,x,t), \tag{11}$$

Here $c_{\text{eff}} \le v_{\max}$ (from (5)), Γ is the phase-noise correlation rate (estimate from the temporal autocovariance of δ_t), and $\lambda_s = \ell_0\sqrt{s}$ is the screening length (Eq. (8)). In the slow limit ($\omega \ll \Gamma$, $v_{\text{flow}} \ll c_{\text{eff}}$), (11) reduces to the Poisson mapping in §6.4. For fast transients (merger-like bursts), $[\partial_t^2 + 2\Gamma\partial_t]$ captures retardation and damping limited by $v_{\max}$.
How to estimate $(c_{\text{eff}}, \Gamma, \kappa_d)$. (i) c_{eff} from spatio-temporal cross-correlations and the bound $c_{\text{eff}} \le a\,\sup_{\mathbf{q}}\|\nabla_{\mathbf{q}}\varepsilon\|/(1+s\Lambda_{\max}^2)$ (Eq. (5)). (ii) $\Gamma \approx D_{\text{phys}}$ from the demodulated phase process ($D = \frac{1}{2}S_\delta(0)$ per step; convert to s^{-1} using τ). (iii) κ_d by least-squares fitting of (11) on held-out data.

7 Experiments and Estimators (Reproducible Recipes)

A/B: PBC vs APBC (operational flip)

Fit AR(2) on identical streams under PBC and APBC; report $(\widehat{V}_L, \widehat{Q}_L)$ and the demodulated $(\widehat{\rho}, \widehat{D})$. Expect a sign flip of $\widehat{V}_L$; the intrinsic V_L is unchanged.

Tail and decoherence sweep

Synthetically vary tail power to scan ρ and D; compare empirical $|\mathcal{C}(t)|$ to the prediction $|\widehat{r}_+|^t e^{-\widehat{D}t}$.

Finite-loop calibration

On a grid with known curvature, vary h, ℓ, and s to measure each contribution in (6) (log–log slopes).

Dynamic pulse test (telegraph validation)

Inject a localized, time-varying innovation; measure $S_\delta(0, x, t)$ and the induced $\Phi(x, t)$. Fit $(c_{\text{eff}}, \Gamma, \kappa_d)$ from (11); check $c_{\text{eff}} \leq v_{\max}$ and Poisson recovery when $\omega \ll \Gamma$.

Minimal estimator (Python-like pseudocode).

```
# Fit AR(2): y_{t+2} = V y_{t+1} - Q y_t + u_t
# Get r_plus, demodulate, estimate D = 0.5 * S_delta(0)
```

A Details: CH Proof, Ring Green's Function, and Phase-Diffusion Estimator

A. CH proof on M^ℓ

Explicit matrices, recursion for M^ℓ, and projection show (1).

B. Ring Green's function

Fourier synthesis on $\mathbb{Z}/L\mathbb{Z}$ yields $G_n^{(L)}$ and $\sum G_n^{(L)} = 0$ (net-zero).

C. Robust $S_\delta(0)$ at zero frequency

Use Bartlett weights $w_k = 1 - k/(W+1)$ on autocovariances $\gamma(k)$ up to lag W: $S_\delta(0) = \gamma(0) + 2\sum_{k=1}^{W} w_k\, \gamma(k)$, with amplitude-gated phase increments.

B Sketches: Strang+BCH, non-Abelian Stokes, and Poincaré/trace

We bound nested commutators (order s), surface terms via F, and boundary integrals by L^2 energies.

C Directed graphs: safe α and singular-value radii

Choose $\alpha \in (0, \|\mathsf{A}\|_2^{-1})$; replace eigenvalues by singular values in all spectral bounds and estimators.